WPM for AI-System Prediction

A mathematical study on training progress, transitions, oscillation, instability and failure

Thy Nguyen

29 October 2026
Mathematical methods manuscript, revised edition
DOI: 10.5281/zenodo.23784369

Abstract

This paper develops a target-conditioned monitoring framework for AI training and other instrumented AI systems. It defines location, scale, progress, regime-transition, oscillation, instability, plateau and failure targets; constructs causal resolution landscapes; and explains when each prediction is mathematically appropriate. A conditional scaling-spectrum representation can provide structural features when its moments and fit are admissible. Exact quadratic-training examples derive convergence, oscillation and instability conditions, while probability and survival calculations distinguish endpoint risk from arrival risk. Application rules address censoring, future schedules, interventions and transfer across architectures without assuming that one telemetry representation predicts every kind of AI behavior.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

Scenario-specific companion papers. The methods in this overview are expanded into separate manuscripts on structural transition, oscillation, instability, plateau transition, foreseeable failure and optimization response. Those manuscripts keep their target definitions, assumptions and telemetry comparisons separate.

Evidence interpretation. The AI predictor is learned from histories, not fed an optimizer's true future equation. Archived comparisons are target dependent: compact structural features

were stronger than telemetry for some transition and plateau targets, similar in oscillation ranking at a longer horizon, and weaker for gradient-defined instability. In a separate latent-failure comparison, long-horizon location-plus-scale AUROC was close to telemetry (0.822 versus 0.819), while other horizons and a later cohort differed [7, 8]. This is not a universal equivalence claim. The compact multi-system scenario source used location/scale proxies and did not compute the complete n/s spectrum.

1 What an AI-system forecast means

The intended reader knows regression, probability, basic optimization and neural-network training terminology. An AI system exposes measurable processes: training and validation loss, gradient norms, parameter-update norms, learning rates, throughput, latency, resource use and evaluation outcomes. A forecasting method needs an observable target and a sampling clock. An iteration, token, batch, wall-clock second and optimizer step are not interchangeable time units.

Let the causal telemetry vector be

$$X_t = (L_t^{\text{train}}, L_t^{\text{val}}, \|g_t\|, \|\Delta\theta_t\|, \eta_t, B_t, C_t), \tag{1.1}$$

where η_t is learning rate, B_t batch size and C_t verified context such as architecture, optimizer and data-policy state. Missing telemetry is marked missing. Validation measurements belong to $\mathcal{F}_t$ only after their actual reporting time, even if associated with an earlier checkpoint.

WPM organizes representation and forecast choice through target T, horizon P, history W and internal resolution m. It does not replace the target predictor with a universal source equation. Suitable targets are repeated measurable outcomes with enough relevant historical examples or a credible supplied dynamical model. Open-ended semantic correctness, arbitrary future task success and unforeseen external failures are not determined by pocket geometry alone.

2 A structural representation inherited from the released paper

When generalized scaling spectra are meaningful, the released n/s family [2] can supply causal descriptors. For moment order $q \neq 0$,

$$H(q,t) = \frac{|q|^{n(t)-1}}{s_0(t)e^{k(t)q}}, \qquad s_0(t) > 0. \tag{2.1}$$

Writing $z_q = \log H(q,t)$ gives the linear parameterization

$$z_q = (n-1)\log|q| - \log s_0 - kq. \tag{2.2}$$

The descriptor is $\Theta_t = (n_t, \log s_{0,t}, k_t)$, possibly accompanied by fit error, conditioning and admissibility flags. The resolution parameter m used for temporal sampling is distinct from moment order q and from any moment-order cutoff.

Proposition 2.1 (Conditional descriptor identification)**.** *If valid positive spectrum values are supplied at moment orders whose design columns* $(\log|q|, 1, q)$ *have full rank, the noiseless log-spectrum parameters are unique within this family.*

Proof. The difference of two fits lies in the nullspace of that design. Full column rank makes the difference zero. Convert the fitted intercept b to $s_0 = e^{-b}$ and the q coefficient c to $k = -c$. □

Worked example 2.2 (A transparent spectrum trail)**.** Take $n = 1$, $s_0 = 2$, $k = 0.1$. Then $H(q) = \frac{1}{2}e^{-0.1q}$. The equal-magnitude ratio is $H(-q)/H(q) = e^{0.2q}$, so $k = \log[H(-q)/H(q)]/(2q)$. At $q = 1$ this gives k directly. A second moment magnitude helps separate curvature from level. These are constructed spectrum values, not measured training results.

Negative moments require care near zeros; high moments can be unstable or nonexistent for heavy-tailed observables. Clipping or adding an offset changes the object analyzed. A conditional fit is not permission to extrapolate to arbitrary moment orders. Fit on admissible observed orders and assess predicted additional orders before interpreting parameters outside that range. A fit to a spectrum is also not, by itself, a forecast of training loss.

3 Target-specific lenses and causal pockets

Let $T \in \{\text{location}, \text{scale}, \text{progress}, \text{transition}, \text{oscillation}, \text{instability}, \text{plateau}, \text{failure}\}$. For each T construct

$$\widehat{L}_t(r; T, P) = \frac{\sum_{s:\, s+P\le t} w_{t,s}\ell_T(Y^T_{s,P}, \widehat{Y}^T_{s,P,r})}{\sum_{s:\, s+P\le t} w_{t,s}}, \quad r = (W, m). \tag{3.1}$$

The labels in this display are assumed to require no follow-up beyond $s+P$; extend their maturity time when a persistence requirement or evaluation delay demands it. Unobserved outcomes never enter as negatives.

Extract connected low-risk regions using a declared grid adjacency and threshold. Retain each pocket's flow curves, position and existence, signed responses, support and identity confidence. A short window may capture a new transition while a long window supports stable trend estimation. Multiple pockets can encode that ambiguity without implying several physical training mechanisms.

Proposition 3.1 (Different losses can select different lenses)**.** *There is no general requirement that a location-optimal lens is scale-optimal or event-optimal.*

Proof. The objectives are different functions. For a counterexample, let two deterministic location predictors have squared errors 0 and 1, while their binary-event probabilities for an event known to occur are 0 and 1. The first minimizes location error and the second minimizes Brier error. No identity links the two candidate orderings without additional assumptions. □

The same argument applies to bias, noise and different target semantics. Estimate separate surfaces or explicitly justify a shared objective.

4 Scenario I: future location and expected progress

For an observable Z_t such as validation loss or a structural coordinate, define location $\mu_{t,P} = \mathbb{E}[Z_{t+P} \mid \mathcal{F}_t]$. A predictor may use telemetry, descriptors and pocket history:

$$\widehat{\mu}_{t,P} = f_\theta(X_{\le t}, \Theta_{\le t}, \mathcal{P}_{\le t}, P). \tag{4.1}$$

The function and its parameters must be supplied or fitted using permitted observations. For progress define $\Delta_P = L_t - L_{t+P}$; positive values mean loss improvement. Then $\mathbb{E}[\Delta_P \mid \mathcal{F}_t] = L_t - \mathbb{E}[L_{t+P} \mid \mathcal{F}_t]$ if L_t is already known. A probability of useful improvement is a separate target $\mathbb{P}(\Delta_P > \delta \mid \mathcal{F}_t)$, with threshold δ fixed before use.

Worked example 4.1 (Prediction versus a decision to continue)**.** Suppose a proposed predictive law is $L_{t+P} \sim N(0.30, 0.02^2)$ and current loss is 0.35. Expected improvement is 0.05. The probability of improvement exceeding 0.03 is $\mathbb{P}(L_{t+P} < 0.32) = \Phi(1)$. This probability assumes the supplied law; expected improvement alone does not determine it. A decision to continue training also requires the cost of P additional steps and the value of improvement.

For throughput or latency, a point prediction may not answer a service-level question. Tail probability and interval-maximum targets should be defined separately, especially when rare stalls dominate operational cost.

5 Scenario II: forecast scale and multivariate envelopes

A future scale target may be dispersion of loss increments over a specified interval, conditional variance of a future checkpoint, or a multivariate predictive covariance. Define which. One endpoint covariance is

$$\Sigma_{t,P} = \mathbb{E}[(Z_{t+P} - \mu_{t,P})(Z_{t+P} - \mu_{t,P})^\top \mid \mathcal{F}_t]. \tag{5.1}$$

It is not necessarily the recent measurement-noise covariance. A model can enforce positivity using $\Sigma = LL^\top + \varepsilon I$ with $\varepsilon > 0$.

For a candidate Gaussian law, a quadratic region is $\{z : (z-\mu)^\top \Sigma^{-1}(z-\mu) \le q\}$. It has the desired coverage only under the law and the correct threshold distribution, or another justified calibration construction. Finite-sample uncertainty in estimated μ, Σ changes the nominal Gaussian calculation.

Worked example 5.1 (Boundary risk can rise while the mean remains safe)**.** Let a scalar instability boundary be $Z > b = 1$. If the predictive mean is 0.8, SD 0.1 gives tail probability $1 - \Phi(2)$, whereas SD 0.2 gives $1 - \Phi(1)$. The mean stays below the boundary. This shows why scale can matter, assuming Gaussian predictive errors. Mere intersection of a chosen envelope with the boundary does not specify either tail probability.

6 Scenario III: oscillation and a quadratic training model

Consider exact gradient descent on the scalar quadratic $L(\theta) = \lambda\theta^2/2$, $\lambda > 0$, with fixed learning rate η:

$$\theta_{t+1} = (1 - \eta\lambda)\theta_t = a\theta_t. \tag{6.1}$$

Theorem 6.1 (Convergence, sign oscillation and instability)**.** *For nonzero initial state, convergence to zero occurs exactly when $|a| < 1$. For $0 < \eta\lambda < 1$, the state contracts without sign alternation; for $1 < \eta\lambda < 2$, the state alternates sign while contracting. At $\eta\lambda = 2$ it alternates without contraction. For $\eta\lambda > 2$ its magnitude grows. At $\eta\lambda = 1$, it reaches zero in one step.*

Proof. The exact solution is $\theta_{t+h} = a^h\theta_t$. Magnitude contracts exactly when $|a| < 1$, and the sign alternates when $a < 0$. Substitute $a = 1 - \eta\lambda$ to obtain the intervals and endpoints. □

Loss follows $L_{t+h} = a^{2h}L_t$. Consequently sign-oscillating parameters can coexist with monotonically decreasing loss. A monitor using loss alone may not observe the oscillation the target intends to detect.

For a general telemetry signal, define oscillation mathematically: sustained negative lag-one correlation, a spectral-band energy ratio above a threshold, or repeated sign reversals of an

observable. These are different labels. A spectral label computed over $t+1:t+P$ is a future target, never an issuance feature. Trend and periodic learning-rate schedules can produce oscillatory telemetry without indicating divergence.

7 Scenario IV: instability and finite-horizon boundary crossing

Let Z_t be a monitored instability statistic such as gradient norm or a rolling range. With a declared past span w, define $R_t = \max_{t-w\le s\le t} Z_s - \min_{t-w\le s\le t} Z_s$. An issuance threshold b_t can be fixed over the forecast, giving arrival time

$$\tau_t = \inf\{h \ge 1 : R_{t+h} > b_t\}. \tag{7.1}$$

Specify whether a crossing must persist, whether a missing measurement censors it, and whether the rolling window includes observed pre-issuance values. The target $\mathbb{P}(\tau_t \le P \mid \mathcal{F}_t)$ differs from $\mathbb{P}(R_{t+P} > b_t \mid \mathcal{F}_t)$.

Worked example 7.1 (A path can cross and return)**.** Suppose the next two states are $(1.2, 0.7)$ with probability $1/2$ and $(0.7, 0.7)$ with probability $1/2$, with boundary 1. The endpoint exceedance probability at step 2 is zero, while arrival probability by step 2 is $1/2$. An endpoint-only model misses this event by definition.

For the unstable scalar quadratic with $|a| > 1$, exact noiseless magnitude crosses a boundary b after the first integer h satisfying $|a|^h|\theta_t| > b$. When $b \ge |\theta_t|$, this is $\lfloor \log(b/|\theta_t|)/\log|a| \rfloor + 1$. Noise and parameter uncertainty require a distribution over crossing times, not this deterministic formula with a fitted coefficient treated as exact.

8 Scenario V: recognition-gated plateau prediction

A plateau may mean small improvement, a small gradient, or failure to improve a validation objective. These need not agree. Define, for example,

$$E^{\text{plat}}_{t,P} = \mathbf{1}\{\text{there exists } h \le P \text{ such that } |L_{t+h+j} - L_{t+h+j-1}| \le \epsilon \text{ for } j = 1, \dots, d\}. \tag{8.1}$$

This definition needs follow-up through as late as $t+P+d$, not merely $t+P$. Use a shorter allowed onset interval if follow-up must end at $t+P$.

An applicability gate $R_t \in \{0, 1\}$ may accept situations where a plateau model is meaningful. Report

$$p^{\text{plat}}_t = \mathbb{P}(E^{\text{plat}}_{t,P} = 1 \mid \mathcal{F}_t, R_t = 1) \tag{8.2}$$

only on accepted cases, with abstention elsewhere. If R_t is already a function of $\mathcal{F}_t$, its presence does not add information; it declares a reporting subset. A conditional model trained on selected cases may not apply to rejected cases. Always retain acceptance coverage as part of the output.

Worked example 8.1 (A loss-improvement plateau in exact contraction)**.** If $L_{t+h} = L_t\rho^h$ with $0 < \rho < 1$, one-step improvement at offset h is $L_t(1-\rho)\rho^h$. It falls below ϵ when

$$h \ge \max\left(0, \left\lceil \frac{\log[\epsilon/(L_t(1-\rho))]}{\log\rho} \right\rceil\right). \tag{8.3}$$

This assumes the nonstrict criterion and no noise. It predicts small absolute improvement, not a poor optimum: even a healthy convergent run eventually meets it. If $\rho = 0$ or $\rho = 1$, handle those cases directly rather than dividing by $\log\rho$.

9 Scenario VI: regime transition and failure

A transition target asks whether a declared regime relation ceases to hold. A failure target asks whether a meaningful failure condition is reached. A scheduled learning-rate change is a regime change but not automatically a failure. High training noise can be compatible with improving generalization; low training loss can coexist with validation deterioration.

Define a failure region $\mathcal{B}$ in monitored state space and first-entry time $\tau = \inf\{h \geq 1 : Z_{t+h} \in \mathcal{B}\}$. The cumulative risk is $C_t(P) = \mathbb{P}(\tau \leq P \mid \mathcal{F}_t)$. Conditional hazards $h_t(j) = \mathbb{P}(\tau = j \mid \tau \geq j, \mathcal{F}_t)$ give

$$C_t(P) = 1 - \prod_{j=1}^{P} [1 - h_t(j)]. \tag{9.1}$$

Proof. Apply the chain rule to survival through steps $1, \ldots, P$. Each factor conditions on survival through the previous step. Subtract the survival probability from one. □

No independence of times is assumed. At a fixed issuance, $C_t(P)$ must be nondecreasing in P. Independently fitted endpoint classifiers need not obey this constraint and do not become survival models merely by sorting their outputs.

Worked example 9.1 (A foreseeable-warning horizon)**.** Hazards $(0.1, 0.2, 0.15)$ give cumulative risks $(0.1, 0.28, 0.388)$. On the predeclared grid $P \in \{1, 2, 3\}$, the first horizon with risk at least 0.3 is 3. This is a functional of a coherent arrival distribution, not a claim that an unconstrained optimal horizon has been discovered.

If a run stops before the horizon, its negative outcome is censored. Include only complete windows or use a survival likelihood with at-risk indicators. Stopping due to an intervention may be informative censoring; the inference then requires a model of that policy.

10 From pocket histories to a predictive probability equation

Let d_t summarize causal telemetry and let ϕ_i retain pocket features and histories. A transparent event-model construction is

$$p_t = \operatorname{logit}^{-1}\left(\beta_0 + \beta_d^\top d_t + \sum_i \alpha_i \phi_i + \sum_{i<j} \gamma_{ij} \psi_{ij}\right). \tag{10.1}$$

The sums must be permutation invariant in pocket labels; tied parameters, normalized group summaries or an explicitly defined graph model can ensure this. Pair features can include flow agreement, separation and co-existence, but geometric proximity alone is not evidence of predictive coupling. Parameters are fitted from matured labels, with regularization appropriate to sample size and dependence.

Theorem 10.1 (Why Brier loss targets event probability)**.** *If $Y \in \{0, 1\}$ and $p_* = \mathbb{P}(Y = 1 \mid \mathcal{F}_t)$, then*

$$\mathbb{E}[(Y - p)^2 \mid \mathcal{F}_t] = p_*(1 - p_*) + (p - p_*)^2. \tag{10.2}$$

Proof. Expand $p_*(1 - p)^2 + (1 - p_*)p^2$ and complete the square. The unique minimizer over $p \in [0, 1]$ is p_*. □

A logistic output lies in $[0, 1]$, but the theorem does not say a fitted logistic model has learned p_*. Proper scoring and calibration are distinct from structural ranking [4]. Strictly monotone recalibration preserves score order while changing probability scale; it need not improve every subgroup.

11 Transfer across AI systems and effects of intervention

Architecture, optimizer, batch size, dataset and hardware can change both the telemetry law and target meaning. Normalize quantities with a stated rationale, such as update-to-parameter ratios or step-to-token conversion. Keep system normalization distinct from the event threshold. A fixed gradient-norm threshold across differently parameterized networks may not define the same failure event.

Partial compatibility means retaining qualified shared components while withholding unsupported ones. A source that supplies a useful throughput model need not supply a valid loss-instability model. If a total resource target adds components, all required terms need coverage; do not average different terms as alternative total estimates.

Known future schedules can be used when they were fixed and available at issuance. Unknown future data order, adaptive optimizer changes or externally chosen actions must be forecast or conditioned on as scenarios. For $X_{t+1} = f(X_t, u_t) + \epsilon_t$, a predictive law under the current policy is generally different from the law under $\mathrm{do}(u_t = u)$. Correlation between a warning and a later learning-rate reduction does not identify the effect of that reduction.

An alert changes the monitored process if it triggers early stopping or remediation. State whether the forecast targets the observed policy or a hypothetical unchanged policy. Historical pocket reliability under one policy may not transport after the policy changes.

12 A multivariate and noisy training equation trail

For quadratic loss $L(\theta) = \frac{1}{2}\theta^\top H\theta$ with symmetric positive definite H, fixed-step gradient descent is $\theta_{t+1} = (I - \eta H)\theta_t$. Write $H = U\Lambda U^\top$, $\Lambda = \mathrm{diag}(\lambda_i)$, and $z = U^\top\theta$. Then each mode follows $z_{i,t+1} = (1 - \eta\lambda_i)z_{i,t}$.

Theorem 12.1 (Multivariate quadratic stability)**.** *The deterministic iteration converges to zero for every initial state exactly when $0 < \eta < 2/\lambda_{\max}$. A mode changes sign each step when $\eta\lambda_i > 1$. Modes with initially zero coefficients do not reveal their stability through that particular trajectory.*

Proof. Orthogonal diagonalization reduces the recurrence to independent scalar recurrences. Convergence for every state requires $|1 - \eta\lambda_i| < 1$ for every eigenvalue. Their common interval is the stated condition. Sign follows from the sign of each scalar multiplier. □

This theorem applies to an exact positive-definite quadratic with fixed learning rate. A non-quadratic neural loss has a state-dependent Hessian, optimizer memory, stochastic gradients and changing data. Local quadratic reasoning can guide a model, but replacing all these effects by one fixed $\lambda_{\max}$ is an approximation whose horizon must be stated.

Now add independent zero-mean scalar gradient noise ξ_t with variance σ_ξ^2:

$$\theta_{t+1} = a\theta_t - \eta\xi_t, \qquad a = 1 - \eta\lambda. \tag{12.1}$$

Conditional on the current state and with future noises independent of it,

$$\mathbb{E}[\theta_{t+h} \mid \theta_t] = a^h \theta_t, \tag{12.2}$$

$$\mathrm{Var}(\theta_{t+h} \mid \theta_t) = \eta^2 \sigma_\xi^2 \sum_{j=0}^{h-1} a^{2j} = \eta^2 \sigma_\xi^2 \frac{1 - a^{2h}}{1 - a^2}, \qquad a^2 \neq 1. \tag{12.3}$$

The first equality follows by iterating the recurrence; cross terms in the variance vanish by independence. If $|a| < 1$, stationary variance is $\eta^2 \sigma_\xi^2 / (1 - a^2)$. At $a^2 = 1$, the finite-horizon variance is $h\eta^2 \sigma_\xi^2$ rather than the displayed fraction.

The expected future quadratic loss is

$$\mathbb{E}[L_{t+h} \mid \theta_t] = \frac{\lambda}{2} \left[a^{2h} \theta_t^2 + \eta^2 \sigma_\xi^2 \sum_{j=0}^{h-1} a^{2j} \right]. \tag{12.4}$$

This has a noise floor even while the expected state contracts. A small mean trajectory does not imply small expected loss or low failure risk. Under Gaussian gradient noise, the future scalar state is Gaussian and a two-sided boundary $|\theta| > b$ has probability

$$\Phi\left(\frac{-b - \mu_h}{s_h} \right) + 1 - \Phi\left(\frac{b - \mu_h}{s_h} \right), \tag{12.5}$$

with the derived μ_h, s_h. This is endpoint risk; first-crossing risk needs the joint recurrence across times.

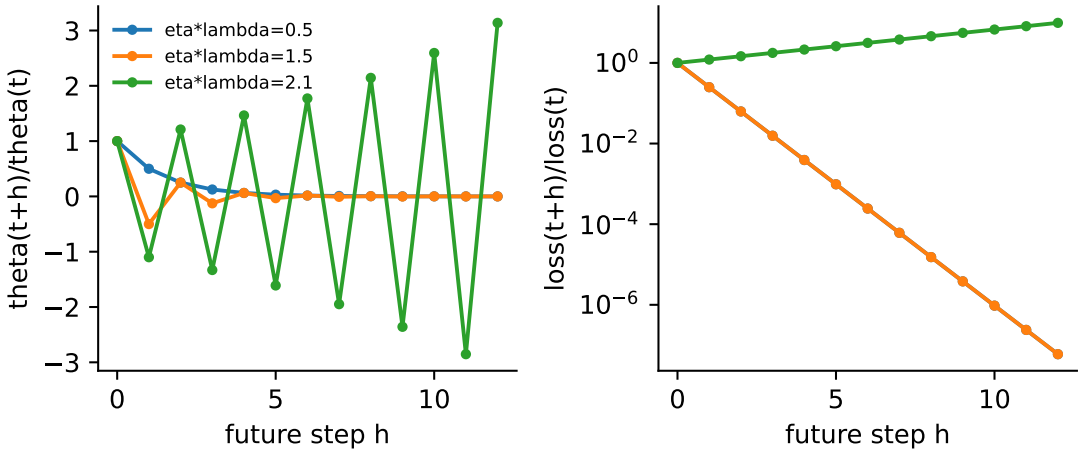


Figure 1: Exact noiseless quadratic dynamics. At $\eta\lambda = 0.5$ parameters contract without alternating; at 1.5 they alternate while loss contracts identically; at 2.1 both magnitude and loss grow. A target's observability depends on the monitored signal.

13 An end-to-end AI-monitoring worksheet

Suppose the query is: "Will a gradient-norm boundary be crossed in the next 20 optimizer steps under the already announced learning-rate schedule?" This fixes the observable, clock, endpoint and policy. It is not a question about semantic output quality or the next 20 wall-clock seconds. Set a boundary and persistence rule, then construct complete-window labels or a censored survival dataset.

At each historical issuance, fit candidate predictors using only the available telemetry and optional admissible n/s descriptors. Enumerate W and m independently, for example to compare fast local variation with a longer structural baseline. Store each issued event forecast. Only after its label matures can its Brier or log loss enter the corresponding landscape. Extract pockets from that surface, retaining full flow, position, existence and identity confidence.

At the current issuance, a qualified location model might supply a predicted state, a scale model a positive uncertainty level, and a hazard model an arrival distribution. These can be separate fitted functions. A general candidate event equation is the logistic construction above, but coefficients must be estimated; a pocket depth cannot be inserted as a probability without a model. If several compatible lenses forecast the same event, combine probabilities with nonnegative weights or fit an explicitly declared distributional aggregation method.

Next apply any fixed calibrator learned from separate permitted labels. State whether its reliability is marginal across runs, within a target horizon, or conditional on an accepted subset. A small-sample or shifted-context situation may cause abstention. The monitor then outputs the event definition, horizon, predicted probability, model assumptions, applicability status and action threshold. If an intervention follows, the next issuance is under a changed policy and must be treated accordingly.

The same construction adapts to progress, plateau, oscillation or latency by replacing the target, label and loss. It should not reuse one event's probability calibration for another merely because they share a telemetry history. Transfer across architectures additionally requires comparable normalization, target semantics and conditional behavior.

14 Warnings, resource decisions and scientifically interpretable use

An operational warning policy should distinguish the cost of waiting, the cost of a false alert and the effect of the contemplated action. With a single binary event and false-alert and miss costs, the familiar probability threshold follows expected-cost comparison. For repeated decisions, a sequential policy can depend on elapsed warning duration and remaining budget; the single-decision threshold need not remain optimal.

Compute-budget prediction is another suitable measurable task when its inputs are available. For a fixed remaining workload N and predicted throughput v, using $N/\mathbb{E}[v]$ as expected time can be wrong because $\mathbb{E}[1/v] \neq 1/\mathbb{E}[v]$. A distribution over throughput and stalls should be propagated to completion time. For model quality, a fixed evaluation set defines a measurable outcome, but repeated reuse can adapt the system to that set. The prediction target then changes with the evaluation policy.

WPM can organize these problems through target-specific resolution and compatible pocket histories. The mathematical advantage is an explicit language for what is observed, what is predicted, and under which assumptions. Whether a particular implementation improves an AI monitor is a separate empirical question; no theorem in this study converts a pocket's appearance into guaranteed future behavior.

15 A deployment procedure

1. Define the monitored observable, clock, reporting delay, target event and horizon. State policy assumptions and what actions are known at issuance.

2. Keep a complete causal ledger of telemetry and issued forecasts. Mark missing or delayed validation outcomes.
3. If using n/s descriptors, check moment admissibility, spectrum-fit rank and the range of meaningful moment orders. Retain failed-fit indicators.
4. Construct separate W,m landscapes for location, scale and each event. Keep independent regimes and individual pocket records.
5. Start from an explicit reference predictor. Combine eligible same-target candidates using Fixed, Multi or a declared Auto policy; do not assume the most elaborate policy is necessary.
6. Fit location, scale, hazard or event parameters from matured labels. Use separate information for tuning and calibration; incomplete follow-up is censored.
7. Return forecast, uncertainty interpretation, applicability status and horizon. For gated plateau forecasts also return acceptance coverage.
8. Refresh only when new information arrives. After an intervention, state whether the target policy and model assumptions remain applicable.

For reliability assessment, distinct runs or genuinely independent episodes are usually more defensible units than overlapping checkpoints. Overlapping labels share data. Reusing future validation measurements in current descriptors is leakage even if their checkpoint time is earlier. Split-conformal guarantees need exchangeability or another justified extension, not merely chronological storage [5].

16 Scenario selection and solved exercises

Question	Suitable mathematical output	Main condition
Where will loss or state be?	Conditional location and distribution	Stable target definition and usable historical or modeled relation
How variable will it be?	Target-specific scale or covariance	Predictive dispersion distinguished from measurement noise
Will improvement be useful?	Distribution of $L_t - L_{t+P}$	Fixed improvement threshold and cost
Will oscillation begin?	Probability of a declared signal criterion	Observable capable of seeing the intended oscillation
Will instability arrive?	First-entry risk or hazard	Boundary, persistence and follow-up rules
Will progress plateau?	Recognition-gated event probability	Explicit plateau semantics and acceptance coverage
Will training fail?	Failure-region arrival probability	Comparable failure meaning and intervention policy
Will service latency breach?	Tail or interval-maximum probability	Resource context and temporal dependence

These are mathematically suitable applications, not a ranking of demonstrated empirical strengths. Each needs its own target predictor and qualification.

Exercise 16.1. For quadratic training let $\eta\lambda = 1.5$. Does loss oscillate in sign? Do parameters alternate?

Solution. $a = -0.5$, so parameters alternate and contract. Loss is nonnegative and multiplies by $a^2 = 0.25$ each step; it decreases without sign oscillation. A loss-only oscillation label would not capture parameter alternation.

Exercise 16.2. A plateau label permits onset by $t + 20$ and requires 5 subsequent confirming increments. At what time is every negative label mature?

Solution. At $t + 25$ under the stated indexing convention. A run observed only through $t + 20$ has insufficient follow-up to rule out a late qualifying onset.

Exercise 16.3. An applicability gate accepts 40% of runs and is accurate on accepted cases. What is still needed to describe its usefulness?

Solution. State the target-specific performance on that accepted population, acceptance coverage, rejection behavior and whether the accepted population transports. Accuracy on accepted cases alone says nothing about the rejected 60%.

Exercise 16.4. An alert causes a learning-rate reduction. Can the later healthy outcome prove that the alert was a false positive for the no-intervention failure target?

Solution. No. The observed outcome is under the intervention policy. The unobserved no-intervention outcome is counterfactual and requires additional causal assumptions or design to identify.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study shares the AI prediction collection DOI 10.5281/zenodo.23784369.

References

[1] T. Nguyen. *The WPM Framework: A Generalized Mathematical Architecture for Cross-Scale, Spatial, and Dynamical Prediction*. Zenodo, 2026. doi:10.5281/zenodo.23248983.

[2] T. Nguyen. *An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation*. Zenodo, 2026. doi:10.5281/zenodo.23111783.

[3] S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*. Cambridge University Press, 2018. https://stanford.edu/~boyd/vmls/.

[4] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[5] G. Shafer and V. Vovk. *Journal of Machine Learning Research* 9, 371–421, 2008. https://jmlr.org/papers/v9/shafer08a.html.

[6] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. https://stanford.edu/~boyd/cvxbook/.

[7] T. Nguyen. *Target-Conditioned Structural Forecasting of AI-Training Transitions: Frozen Multi-System Evidence for 1A–1D*. Version 1.1, October 2026. Archived manuscript, pp. 1–7. Evidence cited only for the scoped comparison statement.

[8] T. Nguyen. *Target-Conditioned Temporal Resolution for AI Training: A Framework for Location, Scale, Regime Transition, and Foreseeable Failure*. Version 3, October 2026. Archived manuscript.